Hysterosalpingographic features of cervical abnormalities: acquired structural anomalies.

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Cervical abnormalities may be congenital or acquired. Congenital cervical structural anomalies are relatively uncommon, whereas acquired cervical abnormalities are commonly seen in gynaecology clinics. Acquired abnormalities of the cervix can cause cervical factor infertility and recurrent spontaneous abortion. Various imaging tools have been used for evaluation of the uterine cavity and fallopian tubes. Hysterosalpingography (HSG) is a quick and minimally invasive tool for evaluation of infertility that facilitates visualization of the inner surfaces of the uterine cavity and fallopian tubes, as well as the cervical canal and isthmus. The lesions of the uterine cervix show various imaging manifestations on HSG such as narrowing, dilatation, filling defects, irregularities and diverticular projections. This pictorial review describes and illustrates the hysterosalpingographic appearances of normal variants and acquired structural abnormalities of the cervix. Accurate diagnosis of such cases is considered essential for optimal treatment. The pathological findings and radiopathological correlation will be briefly discussed.

The role of hysterosalpingography in the evaluation of infertility.

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Hysterosalpingography is the only radiologic procedure routinely performed in the initial evaluation of the infertile woman. Hysterosalpingography is used to assess the anatomy of the uterus and the patency of the fallopian tubes, and is performed in the proliferative phase of the menstrual cycle. It can be performed with either water- or oil-soluble contrast medium. Care should be taken during the procedure that excessive amounts of contrast medium are not injected, because that could obscure the diagnostic findings. Selective salpingography can help evaluate a suspected proximal tubal occlusion. The complications associated with hysterosalpingography include pain, pelvic infection, intravasation of contrast medium and allergic reactions. Abnormal hysterosalpingographic findings include occlusion of one or both fallopian tubes, uterine filling defects and müllerian anomalies. Therapy for abnormal hysterosalpingographic findings depends on the specific clinical scenario and includes endoscopic surgery, laparotomy and in vitro fertilization.

Hysterosalpingography.

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A brief review of the technique of hysterosalpingography is provided. It remains an important diagnostic study in searching for causes of infertility. HSG is a safe, simple procedure that enables the lumina of the uterine cavity and fallopian tubes to be outlined. Fundamentals in techniques have been stressed so as to limit the errors of omission and commission. During the course of many years, I have been honored by my colleagues, who have requested me to review interesting HSGs. Thus, I have been able to accumulate a series of x-rays that I can share with the readers of Modern Trends. Although these photographs seemed unique, many of them probably have been seen by other investigators. They are presented because of their important diagnostic features.

Hysterosalpingography versus sonohysterography for intrauterine abnormalities.

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The hysterosalpingogram is commonly used to evaluate the uterine cavity and the fallopian tubes in the workup of infertile couples. The sonohysterogram is gaining popularity as part of this evaluation. This study compares hysterosalpingography to sonohysterography for the detection of polyps, cavitary fibroids, adhesions, and septae in infertile patients. We conducted a retrospective chart review of 149 infertility patients seen at a University Hospital Center, divisions of Reproductive Endocrinology and Interventional Radiology. Patients underwent hysterosalpingography and sonohysterography as part of their infertility evaluation. The reports were reviewed and findings like polyps, fibroids, adhesions, and septae were compared to the findings obtained at the time of hysteroscopy. Sensitivity, specificity, and accuracy of radiologic tests were the main outcome measures. The sensitivity of hysterosalpingography and sonohysterography was 58.2% and 81.8%, respectively. The specificity for hysterosalpingography and sonohysterography was 25.6% and 93.8%. The differences in sensitivity and specificity were both statistically significant. Hysterosalpingography had a general accuracy of 50.3%, while sonohysterography had a significantly higher accuracy of 75.5%. Although hysterosalpingography is the standard screening test for the diagnosis of tubal infertility and can provide useful information about the uterine cavity, sonohysterography is more sensitive, specific, and accurate in the evaluation of the uterine cavity.

Hysterosalpingographic (HSG) Pattern of Infertility in Women of Reproductive Age.

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Infertility is a complex disorder with significant medical, psychological and economic problems. The aim of the study is to evaluate the structural abnormalities of the uterus and fallopian tubes in infertile women as elucidated by hysterosalpingography. A retrospective study, conducted at the Radiology and Obstetric and Gynaecologic Departments of a tertiary health care institution. Evaluation of all consecutive patients in whom hysterosalpingographic (HSG) was performed for infertility between July 2013 and June 2015 in the Department of Radiology. For the biodata, indications for the investigation and the HSG findings were obtained. The data were analyzed using IBM Statistical Package for the Social Sciences (SPSS Inc., Chicago, IL, USA) for Windows, version 20 software. A total of 299 patients were evaluated. Of these, 250 were for infertility with primary and secondary infertility constituting 18.4 and 81.6%, respectively. Seventy percent of the cases for infertility had abnormalities on the HSG. Normal uterine cavity was found in 123 (49.2%) cases. Uterine filling defects were the most common uterine abnormality. Fallopian tube occlusion, loculated contrast material spillage and hydrosalpinx were more common on the right, and bilateral tubal occlusion was seen only in 11.2%. All cases of intravasation were associated with either unilateral or bilateral fallopian tube blockage or irregularity of the uterus. There was a high incidence of tubal disease in the women presenting with infertility. This was commonly as a result of infection and inflammatory process. This study showed that HSG is very vital in detecting birth canal pathologies; hence, the facility for this important procedure, especially fluoroscopy, should be made available in the health centres for adequate assessment of the women with infertility.

Evaluating Fallopian Tube Patency: What the Radiologist Needs to Know.

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Impaired tubal patency accounts for up to 35% of cases of subfertility and infertility. Hysterosalpingography (HSG) or hysterosalpingo-contrast sonography (HyCoSy) represents a first-line test in evaluating fallopian tube patency. Despite the association of HSG with ionizing radiation, HSG is a reference standard in assessing fallopian tube patency and tubal conditions such as tubal occlusion, salpingitis isthmica nodosa, and hydrosalpinx. HSG is widely available and utilizes either a water-soluble contrast medium (WSCM) or an oil-soluble contrast medium (OSCM). Compared with WSCM, HSG with OSCM results in a higher incidence of non-in vitro fertilization pregnancies and, therefore, may be preferred in women younger than 38 years with unexplained subfertility. HSG may also be helpful in assessment after sterilization or before fallopian tube recanalization. US-based tubal tests are free of ionizing radiation and include HyCoSy, with either air-saline or microbubble US contrast material, and hysterosalpingo-foam sonography (HyFoSy), a tubal patency test that utilizes a gel foam. A comprehensive US infertility evaluation of the pelvis and fallopian tubes can be achieved in one setting by adding coronal three-dimensional imaging of the uterus, saline infusion sonohysterography, and HyCoSy or HyFoSy to routine pelvic US. MR HSG and virtual CT HSG also depict tubal patency and uterine and adnexal pathologic conditions and may be considered in select patients. While laparoscopic chromopertubation remains the standard for tubal patency evaluation, its disadvantages are its invasiveness and cost. Knowledge of the different fallopian tube tests and radiologic appearance of normal and abnormal fallopian tubes results in fewer pitfalls, accurate interpretation, and optimal patient care. Online supplemental material is available for this article. ©RSNA, 2021.

Hysterosalpingography: current applications.

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With the recent advances in reproductive medicine, hysterosalpingography has become a relatively quick and noninvasive examination to evaluate fallopian tubes and uterine cavity. It remains the best modality to image fallopian tubes. Congenital uterine malformations, technical artefacts and pathological findings are depicted. Pathological findings that can be detected on hysterosalpingography include salpingitis isthmica nodosa, tubal blockage, peritubal adhesion, submucosal leiomyoma, endometrial polyp, endometrial carcinoma, synechiae and adenomyosis.

The significance of laparoscopy in determining the optimal management plan for infertile patients with suspected tubal pathology revealed by hysterosalpingography.

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The fallopian tube has numerous functions, including ovum pick-up, the place of fertilization of the ovum and cleavage of the embryo, and transfer of the embryo to the uterus. Tubal pathology impairs functions of the fallopian tube and reduces fertility. The degree of tubal pathology determines the possibility for fertility. The evaluation of the fallopian tube is necessary to determine the management plan of infertility. Hysterosalpingography (HSG) is often performed as a first line approach to assess tubal patency and the presence of adhesions; however, HSG has limitations in detecting tubal pathology. In the current study, we evaluated the significance of laparoscopy in determining the optimal management plan for infertile patients with suspected tubal pathology revealed by HSG. Between 1997 and 2009, 127 patients with suspected tubal pathology as demonstrated by HSG underwent laparoscopy at Kinki University Hospital, and a retrospective analysis was performed. Of 87 patients with unilateral tubal pathology revealed by HSG, 20 patients (23.0%) were given an indication for assisted reproductive technology (ART), based on the laparoscopic findings. Of 40 patients with bilateral tubal pathology revealed by HSG, 33 patients (82.5%) with bilateral tubal pathology detected by laparoscopy were given a high indication for ART. Laparoscopy enables exact evaluation of the fallopian tube and selection of the optimal management plan in infertile patients with suspected tubal pathology revealed by HSG. Therefore, laparoscopy should be performed in infertile patients with suspected tubal pathology revealed by HSG, as it is of diagnostic importance.

Diagnostic Approaches in Nuclear Medicine for Reproductive Health Assessment: Hysterosalpingography in Radiology versus hysterosalpingoscintigraphy.

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Infertility is a significant aspect of reproductive health and evaluating degree of tubal pathology is essential for determining appropriate management plans. To assess the role of hysterosalpingoscintigraphy (HSSG) as a tubal patency test in nuclear medicine and compare it with hysterosalpingography (HSG) in radiology in infertile women and study pain perception in both tests as well. A prospective study was conducted on 50 infertility patients undergoing infertility evaluation at a tertiary care hospital. Both HSG and HSSG procedures were performed during proliferative phase of menstrual cycle. Our study demonstrated the potential of HSSG as a tool for evaluating tubal patency in infertility workup. It showed good accuracy in detecting tubal patency compared to HSG. HSG is a radiological procedure valued for its ability to provide detailed anatomical information of uterus and patency of fallopian tubes. In contrast, HSSG provides dynamic information on the functional aspects of the reproductive system using nuclear medicine techniques. Both HSG and HSSG are vital tools in the diagnostic armamentarium for assessing female reproductive health, offering complementary information that aids in comprehensive patient management.

Comparison of Hysterosalpingography With Laparoscopy in the Diagnosis of Tubal Factor of Female Infertility.

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Background: Laparoscopy is considered to be the gold standard in the evaluation of causes leading to infertility. Hysterosalpingography (HSG) permits indirect visualization of the cervical canal, uterine cavity, and tube patency, which is helpful for evaluating the causes of infertility. Objective: This study aimed to detect tubal abnormalities in infertile women by HSG or laparoscopy and determine the value of HSG in diagnosing fallopian tube status. Methods: The study group consisted of 1,276 patients. HSG was performed as a preliminary test for the evaluation of fallopian tube status. Women were subjected to laparoscopic examination on evidence of HSG abnormalities. Results: The negative predictive value of HSG for detecting patency or occlusion for the right/left tube was 92.08 and 95.44%, respectively. The kappa values for the consistent diagnosis in the right/left tube were 0.470 and 0.574, respectively. In

cases of low patency of the right/left tube, there was a greater than a 40% chance for the tube to be patent, and the remaining high probability was pelvic adhesion. The positive predictive value of HSG for detecting patency or occlusion for both tubes was 87.2%. The kappa value was 0.898 [95% CI (0.838, 0.937), $p < 0.001$], which meant that the diagnostic accuracy of HSG for both tube patency/occlusion was explicit. The kappa value for the diagnosis of hydrosalpinx (especially for bilateral tube hydrosalpinx) was 0.838 [95% CI (0.754, 0.922), $p < 0.001$], and the diagnostic accuracy for HSG was 79.8, 67.9, and 72.4%, respectively. Conclusion: The current study concluded that HSG is a good diagnostic modality to detect tube abnormalities in infertile patients. HSG and laparoscopy are complementary to each other and whenever the patient is undertaken for diagnosis of infertility. Cost-effective HSG had good predictive value in identifying tubal factor infertility.